Theorem 2.2:

Let P be the set of propositional variables, let S be the set of connectives $\{\neg, \wedge, \vee, \rightarrow, \leftrightarrow\}$ and let $v : P \longrightarrow \{T, F\}$ be a function. Then there is a unique truth assignment $\bar{v} : Form(P, S) \longrightarrow \{T, F\}$ such that $\bar{v}(p) = v(p)$ for all $p \in P$. (i.e. $\bar{v} \mid_P = v$, that is $\bar{v}$ restricted to P).

Proof. **Existence** of $\bar{v}$ is established by defining it by recursion on the length of a formula. We exploit the fact that a formula is composed of subformulas of shorter length, with propositional variables as the basic building blocks of length 0.

Consider an arbitrary formula of length 0. Such a formula can only be a propositional variable $p \in P$. So define $\bar{v}(p)$ to be $v(p)$.

Suppose $\bar{v}(\phi)$ has been defined for all formulas of length $\leq n$.

Let ϕ be a formula of length $n+1$. Since $n+1 > 0$ we know that ϕ has a principal connective which is unique. Therefore ϕ is of exactly one of the forms $\neg\theta$, $(\theta \wedge \psi)$, $(\theta \vee \psi)$, $(\theta \rightarrow \psi)$, $(\theta \leftrightarrow \psi)$, with the subformula θ (and ψ) uniquely determined.

Suppose that $\phi = \neg\theta$. Since θ has length $\leq n$, $\bar{v}(\theta)$ has already been defined.

Define: $\bar{v}(\neg\theta) = \begin{cases} T & \text{if } \bar{v}(\theta) = F \\ F & \text{if } \bar{v}(\theta) = T \end{cases}.$

Suppose that ϕ takes on one of the other forms. Since θ and ψ have length $\leq n$, $\bar{v}(\theta)$ and $\bar{v}(\psi)$ have already been defined.

If the principal connective is $\wedge$, define: $\bar{v}((\theta \wedge \psi)) = \begin{cases} T & \text{if } \bar{v}(\theta) = \bar{v}(\psi) = T \\ F & \text{otherwise} \end{cases}.$

If the principal connective is $\vee$, define: $\bar{v}((\theta \vee \psi)) = \begin{cases} F & \text{if } \bar{v}(\theta) = \bar{v}(\psi) = F \\ T & \text{otherwise} \end{cases}.$

If the principal connective is $\rightarrow$, define: $\bar{v}((\theta \rightarrow \psi)) = \begin{cases} F & \text{if } \bar{v}(\theta) = T \text{ and } \bar{v}(\psi) = F \\ T & \text{otherwise} \end{cases}.$

If the principal connective is $\leftrightarrow$, define: $\bar{v}((\theta \leftrightarrow \psi)) = \begin{cases} T & \text{if } \bar{v}(\theta) = \bar{v}(\psi) \\ F & \text{otherwise} \end{cases}.$

Since ϕ is of exactly one of these forms, with θ (and ψ) uniquely determined, exactly one clause applies to ϕ and specifies one value, so $\bar{v}(\phi)$ is well defined. By the recursion principle, this defines $\bar{v}(\phi)$ for every formula ϕ, yielding a function $\bar{v} : Form(P, S) \longrightarrow \{T, F\}$. By the base clause, $\bar{v}(p) = v(p)$ for all $p \in P$; and since each clause above is precisely the condition for respecting the truth table of its connective, $\bar{v}$ is a truth assignment. $\square$

ALTERNATIVE PROOF

Setup. Let P be a set of propositional variables and $S = \{\neg, \wedge, \vee, \to, \leftrightarrow\}$. Throughout, $\Box$ ranges over the binary connectives $\wedge, \vee, \to, \leftrightarrow$. We use unique readability (Section 2.2): every formula in $Form(P, S)$ is of exactly one of the forms (i) a propositional variable $p \in P$, (ii) $\neg\theta$, (iii) $(\theta \Box \psi)$, and in cases (ii) and (iii) the principal connective and the immediate subformulas θ (and ψ) are uniquely determined.

Theorem 1 (Recursion Theorem for $Form(P, S)$)**.** *Let P be countable, let Y be a set, and let $f_0 \colon P \to Y$, $f_\neg \colon Y \to Y$ and, for each binary connective $\Box$, $f_\Box \colon Y \times Y \to Y$ be functions. Then there is a unique function $G \colon Form(P, S) \to Y$ such that*

(R0) $G(p) = f_0(p)$ *for all* $p \in P$;

(R1) $G(\neg\phi) = f_\neg(G(\phi))$ *for all* $\phi \in Form(P, S)$;

(R2) $G((\phi \Box \psi)) = f_\Box(G(\phi), G(\psi))$ *for all* $\phi, \psi \in Form(P, S)$ *and all binary* $\Box$.

Nothing about truth appears here: this is a purely syntactic fact about $Form(P, S)$. Theorem 2.2 is the instance $Y = \{T, F\}$ (Corollary below); other compositional definitions, such as the length function, are other instances (Remark 3).

Lemma 1 (Subformula-respecting enumeration)**.** *If P is nonempty and countable, then there is an enumeration $\phi_0, \phi_1, \phi_2, \dots$ of $Form(P, S)$, without repetitions, in which every formula occurs after its immediate subformulas; that is,*

(E1) if $\phi_n = \neg\phi_i$ *then* $i < n$;

(E2) if $\phi_n = (\phi_i \Box \phi_j)$ *then* $i, j < n$.

Proof. Formulas are finite strings over the countable alphabet $P \cup S \cup \{(,)\}$, so $Form(P, S)$ is countable; and it is infinite, since $p, \neg p, \neg\neg p, \dots$ are distinct formulas for any $p \in P$. Fix any enumeration $\psi_0, \psi_1, \psi_2, \dots$ of $Form(P, S)$. Build a new list in stages: at stage k, append those subformulas of ψ_k that have not yet been listed, in order of increasing length. Each formula has only finitely many subformulas, so each stage appends finitely many entries; no formula is appended twice; and every formula ψ, being a subformula of itself, is listed by the end of stage k where $\psi = \psi_k$. So the new list is an enumeration of $Form(P, S)$ without repetitions. For (E1) and (E2): suppose χ is appended at stage k and let θ be an immediate subformula of χ. Then θ is a subformula of ψ_k (being a subformula of χ, which is a subformula of ψ_k) of strictly smaller length than χ, so θ was listed at an earlier stage or appended earlier within stage k. Either way θ precedes χ in the list. $\Box$

Proof of the Recursion Theorem. **Existence.** If $P = \varnothing$ then $Form(P, S) = \varnothing$ and the empty function is the required G; so suppose $P \neq \varnothing$ and fix an enumeration $\phi_0, \phi_1, \phi_2, \ldots$ as in the Lemma. Write $D_n = \{\phi_i \mid i < n\}$, so that $D_0 = \varnothing$ and $D_{n+1} = D_n \cup \{\phi_n\}$. By recursion on $n \in \mathbb{N}$ we define functions $G_n \colon D_n \to Y$.

Let $G_0 = \varnothing$, the empty function. Given $G_n \colon D_n \to Y$, let

$$y_n = \begin{cases} f_0(\phi_n) & \text{if } \phi_n \in P, \\ f_\neg(G_n(\phi_i)) & \text{if } \phi_n = \neg\phi_i, \\ f_\Box(G_n(\phi_i), G_n(\phi_j)) & \text{if } \phi_n = (\phi_i \Box \phi_j), \end{cases}$$

and define $G_{n+1} = G_n \cup \{(\phi_n, y_n)\}$.

This is legitimate. By unique readability, exactly one of the three cases applies to ϕ_n, and in the last two the immediate subformulas; hence, as the enumeration has no repetitions, the indices i (and j) — are uniquely determined. By (E1) and (E2) these indices are $< n$, so $\phi_i, \phi_j \in D_n = \mathrm{dom}(G_n)$ and the displayed values exist. Thus y_n is a single, well-defined element of Y. Since $\phi_n \notin D_n$, the set G_{n+1} is a function with domain D_{n+1}, and $G_n \subseteq G_{n+1}$ by construction, so the G_n form a chain.

Let $G = \bigcup_{n \in \mathbb{N}} G_n$. Then:

- G is a function: if $(\phi, y), (\phi, y') \in G$ then both pairs lie in a common G_m, which is a function, so $y = y'$. Moreover each $G_n \subseteq G$, so $G(\phi_i) = G_n(\phi_i)$ whenever $i < n$.
- $\mathrm{dom}(G) = Form(P, S)$: every formula is ϕ_n for some n, hence lies in $D_{n+1} \subseteq \mathrm{dom}(G)$.
- G satisfies (R0)–(R2). For (R0): if $p \in P$ then $p = \phi_n$ for some n, the first case applied at stage n, and so $G(p) = y_n = f_0(p)$. For (R1): given ϕ, we have $\neg\phi = \phi_n$ for some n, and $\phi = \phi_i$ for the unique i given by unique readability, with $i < n$ by (E1); the second case applied at stage n, so $G(\neg\phi) = y_n = f_\neg(G_n(\phi_i)) = f_\neg(G(\phi))$. (R2) is proved in the same way using (E2).

Uniqueness. (No countability of P is needed here.) Suppose $G, H \colon Form(P, S) \to Y$ both satisfy (R0)–(R2), and let

$$X = \{\phi \in Form(P, S) \mid G(\phi) = H(\phi)\}$$

be their agreement set. Then $P \subseteq X$, since$G(p) = f_0(p) = H(p)$ for all $p \in P$ by (R0). X is closed under $\neg$: if $\phi \in X$ then $G(\neg\phi) = f_\neg(G(\phi)) = f_\neg(H(\phi)) = H(\neg\phi)$, so $\neg\phi \in X$. And X is closed under each binary $\Box$: if $\phi, \psi \in X$ then $G((\phi\Box\psi)) = f_\Box(G(\phi), G(\psi)) = f_\Box(H(\phi), H(\psi)) = H((\phi\Box\psi))$, so $(\phi\Box\psi) \in X$. Now $Form(P, S)$ is, by its inductive definition, the smallest set of strings that contains P and is closed under the formula-building operations $\theta \mapsto \neg\theta$ and $(\theta, \psi) \mapsto (\theta\Box\psi)$; and X is such a set, so $Form(P, S) \subseteq X$. Since also $X \subseteq Form(P, S)$, we get $X = Form(P, S)$: the functions G and H have the same domain and agree at every element of it, so $G = H$. $\square$

Corollary 1 (Theorem 2.2)**.** *Let P be a countable set of propositional variables and let $v\colon P \to \{T, F\}$ be a function. Then there is a unique truth assignment $\bar{v}\colon Form(P, S) \to \{T, F\}$ such that $\bar{v}(p) = v(p)$ for all $p \in P$. (Remark 2 removes the countability hypothesis.)*

Proof. Apply the Recursion Theorem with $Y = \{T, F\}$, $f_0 = v$, and the truth-table functions: $f_\neg(T) = F$, $f_\neg(F) = T$, and, for $x, y \in \{T, F\}$,

$$f_\wedge(x, y) = T \iff x = y = T, \qquad f_\vee(x, y) = F \iff x = y = F,$$

$$f_\to(x, y) = F \iff (x, y) = (T, F), \qquad f_\leftrightarrow(x, y) = T \iff x = y.$$

With these choices, a function $u\colon Form(P, S) \to \{T, F\}$ satisfies (R1)–(R2) if and only if u respects the truth tables of all the connectives in S — the equations and the respect-conditions are the same statements read twice, that is, if and only if u is a truth assignment; and u satisfies (R0) if and only if $u(p) = v(p)$ for all $p \in P$. So the unique G produced by the theorem is precisely the unique truth assignment extending v; this is $\bar{v}$. ☐

Remark 1. What the proof achieves is a reduction: recursion on the structure of formulas is reduced to ordinary recursion and induction on $\mathbb{N}$ (the sequence $(G_n)_{n\in\mathbb{N}}$ is itself defined by recursion on n, which we take as basic). The two ingredients doing the real work are unique readability, which makes each y_n single-valued, and the enumeration conditions (E1), (E2), which guarantee that the inputs $G_n(\phi_i)$, $G_n(\phi_j)$ already exist when needed.

Remark 2. Countability. The Lemma requires P to be countable, whereas the book states Theorem 2.2 for arbitrary P. To remove the hypothesis, index by length instead of by list position: let F_n be the set of formulas of length $\leq n$ and prove, by induction on n, that there is exactly one function $g_n\colon F_n \to Y$ satisfying (R0)–(R2) for all formulas in F_n. The restriction of any such g_{n+1} to F_n satisfies the same conditions there (immediate subformulas of a formula in F_n lie in F_n), so equals g_n by the uniqueness conjunct; hence the g_n form a chain and $G = \bigcup_n g_n$ is as required. Note that the uniqueness half of the theorem, proved above by structural induction, already holds for arbitrary P.

Remark 3. Two instances worth recording. Taking $Y = \mathbb{N}$, $f_0(p) = 0$, $f_\neg(m) = m + 1$ and $f_\square(m, k) = m + k + 1$ produces the length function (number of connectives) used throughout. And unwinding the Corollary: extension $v \mapsto \bar{v}$ and restriction $u \mapsto u|_P$ are mutually inverse bijections between the set of all functions $P \to \{T, F\}$ and the set of all truth assignments on $Form(P, S)$.